# Jason Xinyu Liu

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## Academic Appointments

2025 – present **Massachusetts Institute of Technology**
Postdoctoral Associate in Aeronautics and Astronautics
Advisor: Prof. Julie Shah

## Education

2018 – 2025 **Brown University**
Ph.D. in Computer Science
Advisor: Prof. Stefanie Tellex
Dissertation: Robotic Language Grounding [PDF, Video]

2021 **Brown University**
M.S. in Computer Science
Advisor: Prof. Stefanie Tellex

2017 **University of California, Berkeley**
B.S. in Electrical Engineering and Computer Sciences
Advisor: Prof. Ken Goldberg and Prof. David Wagner

## Fellowships, Awards and Honors

Brown **National Science Foundation Graduate Research Fellowship Program**, 2018
**Jack Kent Cooke Foundation Graduate Scholarship**, 2018

UC Berkeley **Tau Beta Pi, National Engineering Honor Society**, 2014
**Eta Kappa Nu, National EECS Honor Society**, 2014
**Jack Kent Cooke Foundation Undergraduate Transfer Scholarship**, 2014

## Publications

CoRL 2026 **Language as a Sensor: Calibrated Spatial Belief Estimation in 3D Scenes from Natural Language**
A. Naveen, **J.X. Liu**, L. Carlone, A. Bobu
*Conference on Robot Learning (CoRL) 2026*. [Paper]

RA-L 2026 **GHOST in the Robots: Real-Time Exocentric Dual-Robot VR Teleoperation from Onboard Cameras**
Y. Wei, F. Zaghloul, S.C. Aryal, A.K. Agrawal, C. Li, **J.X. Liu**, J. Tompkin, S. Tellex
*IEEE Robotics and Automation Letters (RA-L) 2026*. [Paper]

ICRA 2026 **Pragmatic Embodied Spoken Instruction Following in Human-Robot Collaboration with Theory of Mind**
L. Ying, X. Li, S. Aarya, Y. Fang, Y. Yin, **J.X. Liu**, S. Tellex, J.B. Tenenbaum, T. Shu
*International Conference on Robotics and Automation (ICRA) 2026*. [Paper]

HRI 2026 **LEGS-POMDP: Language and Gesture-Guided Object Search in Partially Observable Environments**
I. He, S. Tellex, **J.X. Liu**
*ACM/IEEE International Conference on Human-Robot Interaction (HRI) 2026*. [Paper, Website]

RA-L 2025 **Open-vocabulary Pick and Place via Patch-level Semantic Maps**
M.Jia, H. Huang, Z. Zhang, C. Wang, L. Zhao, D. Wang, **J.X. Liu**, R. Walters , R. Platt , S. Tellex
*IEEE Robotics and Automation Letters (RA-L) 2025*. Best Paper Award Honorable Mention. [Paper, Website]

IROS 2024 **Lang2LTL-2: Grounding Spatiotemporal Navigation Commands Using Large Language and Vision-Language Models**
**J.X. Liu**, A. Shah, G. Konidaris, S. Tellex, and D. Paulius
*IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2024*. [Paper, Website]

IJCAI 2024 **A Survey of Robotic Language Grounding: Tradeoffs between Symbols and Embeddings**
V. Cohen*, **J.X. Liu***, R. Mooney*, S. Tellex*, D. Watkins*
*International Joint Conference on Artificial Intelligence (IJCAI) 2024 Survey Track*. [Paper]

ICRA 2024 **LTL-Transfer: Skill Transfer for Temporally-Extended Task Specifications**
**J.X. Liu***, A. Shah*, E. Rosen, M. Jia, G.D. Konidaris, S. Tellex
*IEEE International Conference on Robotics and Automation (ICRA) 2024*. [Paper, Website]

CoRL 2023 **Lang2LTL: Grounding Complex Natural Language Commands for Temporal Tasks in Unseen Environments**
**J.X. Liu***, Z. Yang*, I. Idrees, S. Liang, B. Schornstein, S. Tellex, A. Shah
*Conference on Robot Learning (CoRL) 2023*. [Paper, Website, Brown News]
*AAAI Fall Symposium on Unifying Representations for Robot Application Development (URRAD) 2023*. Best Paper Finalist.

INTERSPEECH 2023 **Style-transfer based Speech and Audio-visual Scene Understanding for Robot Action Sequence Acquisition from Videos**
C. Hori, P. Peng, D. Harwath, **J.X. Liu**, K. Ota, S. Jain, R. Corcodel, D. Jha, D. Romeres, J. Le Roux
*INTERSPEECH 2023*. [Paper]

ICRA 2022 **Generalizing to New Domains by Mapping Natural Language to Lifted LTL**
E. Hsiung, H. Mehta, J. Chu, **J.X. Liu**, R. Patel, S. Tellex, G. D. Konidaris
*IEEE International Conference on Robotics and Automation (ICRA) 2022*. [Paper]

ACNS 2020 **Minority Reports Defense: Defending Against Adversarial Patches**
M. McCoyd, W. Park, S. Chen, N. Shah, R. Roggenkemper, M. Hwang, **J.X. Liu**, D. Wagner
*Applied Cryptography and Network Security Workshops (ACNS) 2020*. [Paper]

ICRA 2018 **Dex-Net 3.0: Computing Robust Vacuum Suction Grasp Targets in Point Clouds Using a New Analytic Model and Deep Learning**
J. Mahler, M. Matl, **J.X. Liu**, A. Li, D. Gealy, K. Goldberg
*IEEE International Conference on Robotics and Automation (ICRA) 2018*. [Paper, Website]

ICISS 2018 **Detecting Phone Theft Using Machine Learning**
**J.X. Liu**, D. Wagner, S. Egelman
*International Conference on Information Science and System (ICISS) 2018*. [Paper]

RSS 2017 **Dex-Net 2.0: Deep Learning to Plan Robust Grasps with Synthetic Point Clouds and Analytic Grasp Metrics**
J. Mahler, J. Liang, S. Niyaz, M. Laskey, R. Doan, **J.X. Liu**, J. A. Ojea, K. Goldberg
*Robotics: Science and Systems (RSS) 2017*. [Paper, Website]

## Invited Talks

| | |
|---|---|
| October 2026 | CoRL 2-26 Workshop Spatial Intelligence and Structured Reasoning for Embodied Navigation (hosted by Dr. Bhatt, Neel) |
| March 2026 | The University of Edinburgh, RL Seminar (hosted by Dr. David Abel) |
| Oct 2025 | UIUC, CS Colloquium [web] (hosted by Prof. Dilek Hakkani-Tür and Prof. Gokhan Tür) |
| Oct 2025 | Tufts, HRI Seminar (hosted by Prof. Jivko Sinapov) |
| Jan 2025 | USC, Robotics Seminar (hosted by Prof. Jesse Thomason) |
| Dec 2024 | Brown, Brown Database Group (hosted by Prof. Ugur Çetintemel) |
| Oct 2024 | Amazon Robotics, Amazon Robotics Research Fall Symposium and PhD Communication Competition [web] |
| Oct 2024 | UT Austin, Forum for Artificial Intelligence [video] (hosted by Prof. Joydeep Biswas) |
| Oct 2024 | CU Boulder, CS Colloquium [video] (hosted by Prof. Bradley Hayes) |
| Sep 2024 | MIT, Interactive Robotics Group (hosted by Prof. Julie Shah) |
| Sep 2024 | UT Austin, Center of Autonomy (hosted by Prof. Ufuk Topcu) |
| Aug 2024 | Arizona State, Logos Robotics Lab (hosted by Prof. Nakul Gopalan) |
| Jul 2024 | Oxford, CS Department Seminar (hosted by Prof. Giuseppe De Giacomo) |
| Oct 2022 | Northeast Robotics Colloquium |

## Guest Lectures

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|---|---|
| Spring 2026 | Sapienza University of Rome, Hybrid Neurosymbolic Techniques for Non-Markovian Reinforcement Learning |
| Spring 2025 | Brown, CSCI 1470: Deep Learning |
| Fall 2024 | Brown, CSCI 1951X: Formal Proof and Verification |

## Mentorship

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|---|---|
| 2026 | Aryan Naveen (MIT Ph.D.) |
| 2026 | Yiran Tao (MIT Ph.D.) |
| 2026 Summer | Jefferson Lin (MIT Undergraduate) |
| 2026 Summer | Amanda Cheng (Wellesley Undergraduate) |
| 2026 Summer | Jonas Xiangzhou Sun (MIT Undergraduate) |
| 2024 - 2025 | Ivy He (Brown Ph.D.) |
| 2023 - 2025 | Yichen Wei (Brown Master's. Now Ph.D. at UPenn) |
| 2023 - 2024 | Anneke Wernerfelt (Haverford Undergraduate -> Master's at UPenn) |
| Spring 2021 | Shihab Khalfalla (UConn Undergraduate. Now at Microsoft) |
| 2021 - 2022 | Eric Hsiung (Brown Master's. Now Ph.D. at UT Austin) |
| 2020 - 2021 | Tyler Edwards (UPenn Undergraduate -> Master's at UPenn) |
| 2020 - 2021 | Suchen Zheng (Brown Master's. Now Ph.D. at Yale) |

## Workshop Organization

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|---|---|
| CoRL 2026 | Modeling Uncertainty in Robotic World Models. [Website] |
| IROS 2026 | Leveraging Human-Robot Dialogue to Improve Robot Learning and Human-Robot Interaction. [Website] |
| IROS 2026 | Learning and Formal Methods for Robotics. [Website] |

CoRL 2025 Evaluation and Deployment Across the Robot Learning Lifecycle. [Website]

RSS 2025 Robot Planning in the Era of Foundation Models. [Website]

RSS 2024 Task Specification for General-Purpose Intelligent Robots. [Website]

## Service

Brown **First-year Ph.D. student mentor**, 2021-2023
**Google exploreCSR** (explorecsr.cs.brown.edu), Spring 2021

Review **THRI** 2026, **RSS** 2026, **JMLR** 2025, **CoRL-W** 2024, **RSS-W** 2024, **IROS** 2024, **ICRA** 2023, **ICRA** 2022

## Industry Experience

Summer 2022 **Research Intern**, *Mitsubishi Electric Research Laboratories*, Cambridge, MA

Summer 2015 **Software Engineer**, *NetSuite*, San Mateo, CA